FOCUS: Fatigue-reducing IOCs through Prioritization and Scoring

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***Abstract*—"In recent years, the field of Cyber Threat Intelligence (CTI) has undergone rapid development, primarily due to the emergence of various organizations and platforms dedicated to threat intelligence. Through automated analysis and real-time information sharing, companies can now more effectively identify and manage the risks associated with cyber threats and attacks. One of the most commonly used types of threat data is Indicators of Compromise (IOCs), which play a crucial role in early cyber intrusion detection and prevention. However, the quality of the collected data is often limited, and the sheer volume of real-time data can overwhelm security analysts, leading to IOC data fatigue.**

To address this challenge, a new scoring mechanism called FOCUS has been proposed. FOCUS utilizes various attributes and relationships provided by the VirusTotal online scanning engine analysis results to prioritize a set of IOCs. By concentrating on properly analyzed and ranked IOCs from a vast dataset, FOCUS can assist organizations in allocating resources and efforts more effectively. Ultimately, FOCUS can enhance the accuracy and efficiency of IOC analysis, strengthening an organization's ability to respond to cybersecurity threats.

I. Introduction

Indicators of Compromise (IOCs) are forensic artifacts that indicate a security intrusion on a computer or network [1]. Cybersecurity researchers and analysts collect and share IOC data worldwide to analyze it using cyber threat intelligence (CTI) platforms and services, such as VirusTotal online scan engine [2]. This enables better detection and prevention of potential malware attacks and shortens the window between compromise and detection [3].

Examples of IOC data include unusual network traffic, unknown files or processes, suspicious DNS and registry configurations, among others. Although security vendors extract and share most of these IOC data publicly, the insights and relations between IOCs are often limited [4]. Additionally, a study has shown that the quality of most threat intelligence data is unreliable [5].

Moreover, the sheer volume of IOC data makes it challenging for analysts to extract valuable insights from it, leading to data fatigue and prioritization problems [3]. Given the rate at which IOC data is produced, analyzing it with the necessary resources is technically impossible.

Current research trends in IOC mainly focus on the automatic identification and extraction of IOC data using machine learning models and techniques [1], [6]. However, there has been little to no research on addressing the problem of IOC data fatigue and prioritization. This is a significant challenge in the field of cyber threat intelligence [5]. Therefore, our work aims to contribute to resolving the IOC data fatigue problem by developing an effective approach for conducting IOC prioritization and reducing data fatigue, which can help analysts derive actionable insights from these prioritized IOCs.

The primary objective of our work is to develop FOCUS, a scoring mechanism that can effectively prioritize IOCs by analyzing multiple VirusTotal IOC attributes and relationships. FOCUS will provide a prioritization score that can serve as a benchmark for resource allocation and indicate the level of priority assigned to each IOC. For instance, a high prioritization score will signify the need for further examination and monitoring of the IOC. Our goal is to provide analysts with a more effective way to prioritize their workloads and alleviate IOC data fatigue. FOCUS will help analysts make informed decisions and enhance their ability to combat the increasing number of cyber threats.

In our work we will focus solely on the MD5 file hash IOC type. We will analyze and extract further attributes and information of the MD5 file hash IOCs using the VirusTotal online IOC scanning engine API. By limiting the scope to this specific IOC type, we can develop a more targeted and effective scoring mechanism that can accurately assess the level of maliciousness of these IOCs.

A. Problem Statement

The increasing volume of IOCs data is causing data fatigue and challenges for security analysts in effectively prioritizing their investigations and responses. This raises the risk of overlooking critical threats, potentially resulting in data breaches, financial losses, and reputational damage for organizations. Hence, there is a need to explore the factors contributing to IOCs data fatigue, devise strategies for prioritizing IOCs data, and enhance the effectiveness of security analysts in detecting and responding to cyber threats.

1) Research Gaps: While the problem statement addresses IOCs data fatigue, there may be limited research on the

specific contributing factors. Identifying these factors could assist organizations in developing targeted strategies. While the problem statement acknowledges the need to prioritize IOCs data, practical effectiveness may not have been extensively explored. Hence, investigating prioritization and scoring of IoCs and assessing their efficacy could optimize security response efforts.

B. Approach Overview

To generate prioritization scores and rankings for IOCs, FOCUS comprises three main steps:

The first step involves the crawling and processing of IOC data using Python scripts to efficiently manipulate data elements. This step comprises two substeps: ThreatFox IOC Data Crawling and VirusTotal IOC Scanning. Initially, IOCs are crawled from the Threatfox public IOC sharing platform and subsequently analyzed using the VirusTotal online IOC scanning engine. Various VirusTotal IOC data objects can be extracted and examined to provide valuable insights regarding an IOC. The IOC data is collected from monitoring tools and serves as the basis for further processing.

The second step encompasses the extraction and quantification of VirusTotal IOC attributes. In this stage, attributes contributing to the scoring of an IOC are extracted and quantified for use in the 'FOCUS' mechanism. Some examples of extracted attributes include the age of an IOC, the number of IDS alerts, and suspicious command executions.

Finally, the third step involves generating score categories and prioritization results using 'FOCUS,' which utilizes the quantified attributes. Three score categories—base score, suspicious behavior score, and attack techniques score—are computed for each IOC to conduct multiple prioritization experiments.

In our work, two prioritization methodologies have been developed: one-phase prioritization and two-phase prioritization. One-phase prioritization produces prioritization rankings based on the overall prioritization score of IOCs. Conversely, two-phase prioritization involves ranking the IOCs in two phases. In the first phase, the base score of IOCs is calculated and used for ranking. In the second phase, the sum of the suspicious behavior score and attack techniques score is calculated and added to the base score to produce the final rankings.

C. Results Overview

In our work, we have conducted several experiments using FOCUS. We examined the relationship between extracted attributes and score categories to understand their underlying trends and how these attributes contribute to the three score categories. For instance, the age of an IOC is a significant factor contributing to the base score of an IOC.

We visualized the distributions of quantified IOC attributes for the top 100 prioritized IOCs by the one-phase prioritization. This visualization highlights the prioritization patterns and characteristics of FOCUS. For instance, FOCUS tends to prioritize IOCs with a relatively higher number of IDS alerts.

Comparing one-phase and two-phase prioritization, the experiment results indicate a trend where two-phase prioritization assigns higher rankings to IOCs in the high and critical prioritization severity categories. In contrast, IOCs in the low and medium severity categories show a relatively small percentage increase in prioritization rankings.

To validate the characteristics of FOCUS, we incorporated another IOC data source from the ESET research group alongside the existing VirusTotal IOC data source. The experimental results validate and suggest that FOCUS prioritizes IOCs based on several criteria derived from the extracted IOC attributes, aligning with the expected outcome of the experiment.

D. Contributions

Our work aims to make two primary contributions to the field of cybersecurity:

- Introduction of FOCUS, a novel IOC scoring mechanism designed to prioritize IOCs based on their threat level. FOCUS will utilize open threat intelligence platforms such as VirusTotal to extract relevant IOC attributes, which will then be scored to establish a benchmark for prioritization rankings.
- Provision of guidelines for cybersecurity analysts on optimizing their resources when responding to security incidents involving IOCs. By leveraging prioritization rankings as a tool, analysts can channel their efforts toward high-priority threats, potentially reducing response times and minimizing the impact of cyber attacks.

In essence, this paper seeks to offer a practical solution to address the challenges posed by the escalating volume of IOC data and the imperative for effective prioritization strategies in the realm of cybersecurity.

Structure of the Paper: This paper is organized by providing introduction in Section 1. In Section 2, we review the existing related work. Section 3 provides the approach taken by FOCUS. Section 4 provides the methodology and implementation. Section 5 evaluates the performance of FOCUS. In Section 5, we present the results and discussion. Finally, we conclude in Section 7.

II. Related Work

This section offers a brief introduction to the current state of the art in IoC prioritization and scoring.

Quantifying an IOC is crucial in determining its priority. To achieve this, a cyber threat quantification mechanism has been proposed in [7]. This mechanism uses weighted values and a threat level determination matrix to calculate the weighted threat level for each IOC.

In [8], a scoring model is introduced to enhance IOCs detection by amalgamating various open intelligence feeds. It mitigates false positives by considering feed independence to prevent overemphasis on the same data. The model also

incorporates a decay time parameter for assessing IOC relevance over time, favoring recently observed IOCs. Furthermore, it factors in source confidence to acknowledge varying feed reliability. By integrating these aspects, this scoring model offers an improved and precise approach to prioritize and detect genuine positive IOCs.

In [9], a scoring model for IOCs is presented, resembling the one in [8]. Both models use source confidence and decay rates to calculate IOC threat scores. However, [9] introduces extra features, like adjustable end times and customizable decay functions with rates. This flexibility enables analysts to tailor IOC attributes, such as setting end times based on an IOC's expected lifespan or adjusting decay rates to reflect its relevance decline. Thus, the [9] scoring model becomes a versatile tool for precise IOC prioritization and detection.

On the other hand, authors in [10] suggest a method for improving OSINT processing to generate enriched IoCs by correlating and aggregating IoCs from different sources into clusters, thus presenting threat information more effectively. In [11], the authors introduce a methodology for identifying security threats in activity logs. They propose an outlier detection algorithm that employs clustering techniques and an encoding method based on prime numbers to enhance activity analysis. However, both of the above works focus on clustering IoCs rather than prioritization and scoring.

In summary, existing scoring models in related literature aim to assess IOC threat levels. Nevertheless, efficient prioritization of a large number of IOCs requires an accurate and highly configurable scoring mechanism. This allows for precise tuning of hyperparameters to achieve the desired precision and adaptability, tailored to an organization's specific context and threat landscape. Such a scoring mechanism ultimately enhances IOC prioritization, aiding organizations in timely and proactive cyber threat detection and prevention, aligning with the primary motivation and objective of our preliminary work, FOCUS.

III. Our Approach: FOCUS

This section introduces an overview of FOCUS, including its data source and the required data processing steps.

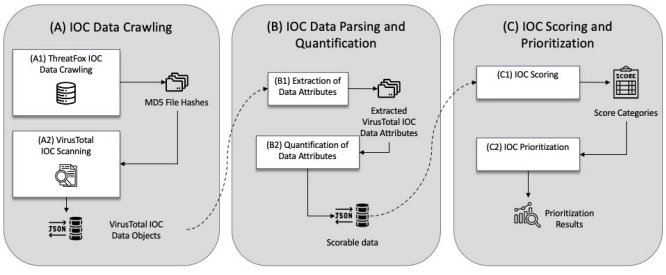


Fig. 1: Overview of FOCUS

A. Overview

To compute a prioritization score for an IOC, our work follows three main steps: (A) IOC Data Crawling, (B) IOC Data Parsing and Quantification, and (C) IOC Scoring and Prioritization. Python scripts are used to crawl and process the VirusTotal IOC data, allowing for efficient and effective manipulation of data elements. Step (A) consists of two substeps: (A1) ThreatFox IOC Data Crawling and (A2) VirusTotal IOC Scanning. In step (B), substeps such as (B1) Extraction of Data Attributes and (B2) Quantification of Data Attributes are performed. Finally, in step (C), the substeps that produce scores and prioritization results, namely (C1) IOC Scoring and (C2) IOC Prioritization, are carried out. The following sections will provide a detailed discussion of these steps and their substeps.

B. IOC Data Crawling

The IOC data used in our work are obtained by crawling the ThreatFox IOC sharing platform, which provides a repository of IOCs shared by various organizations and individuals. These IOCs are then processed in substep (A2) by uploading them to the VirusTotal IOC scanning website, which provides detailed file IOC attributes and analysis results. These attributes and results are extracted to create scores using the analysis data returned by VirusTotal.

1) VirusTotal IOC Scanning: To compute an IOC data score, we must extract valuable data attributes. In our approach, we upload the IOC data, obtained from the ThreatFox sharing platform, to the VirusTotal online malware scanning engine for analysis. We collect and analyze various data attributes from three primary VirusTotal IOC data objects: VirusTotal File object, VirusTotal File Behavior object, and VirusTotal Attack Tactics and Techniques object. These VirusTotal data objects are presented in JSON format and parsed and quantified in step (B).

VirusTotal File Object Primarily consists of fundamental data attributes of an IOC, including threat labels, YARA alerts, SNORT alerts, VirusTotal analysis malicious results, and more.

VirusTotal File Behavior Object This primarily comprises behavioral attributes of an IOC, such as command executions, additions or modifications of registry keys, communication with domains and IP addresses, and more.

VirusTotal Attack Tactics and Techniques Object This object mainly consists of the attack tactics and techniques employed by an IOC, which reference the MITRE ATT&CK knowledge base.

C. IOC Data Extraction and Quantification

After analyzing the various types of collected IOC data attributes from the VirusTotal scanning results, we extract those deemed to significantly contribute to the prioritization score of an IOC in substep (B1) and quantify them into an integer format in substep (B2). Step (B) facilitates the conversion of VirusTotal IOC data objects into a scorable integer format.

Extraction of Data Attributes Substep (B1) involves extracting attributes from deeply nested JSON objects using a Python script. Since the JSON objects contain a large amount of data and the required attributes are deeply nested within the object, extracting them manually would be inefficient and time-consuming. Therefore, the Python script is used to automate the extraction process. The extracted attributes are then passed on to substep (B2), where they are quantified to enable the calculation of the prioritization score.

Quantification of Data Attributes Some attributes in the VirusTotal IOC data object need to be filtered and quantified for scoring in step (C). For instance, IOC command executions are filtered using the examples of suspicious commands and command flags presented in Table V to obtain a count of suspicious commands in integer format. Examples of commands executions found in an IOC:

”command_executions”: [”su”, ”ping -c 2 -W 10 -v google.com”]

D. IOC Scoring and Prioritization

After obtaining quantified data attributes from various VirusTotal data objects, three score categories are abstracted based on the three types of VirusTotal data objects mentioned in subsection III-B1. These score categories are introduced to provide a comprehensive analysis of the IOC and consider different aspects of the data attributes. The following sections will explain each of the indicators that contribute to the prioritization factor of an IOC.

1) Score Categories: Quantified IOC data attributes generated in step (B) will be used to calculate the following three types of score categories. Detailed specifications of the quantified attributes and the mathematical equations for these score categories will be discussed in Section IV. In general, the three score categories will be computed by using the following generalized scoring equation.

$$Score = f(X_1, X_2, ..., X_n)$$

where $X_1, X_2, ..., X_n$ are the quantified VirusTotal IOC data attributes.

a) Base Score: Computed based on the fundamental attributes of a file IOC, which describe basic statistics and information about the analyzed IOC data. For example, the age of an IOC, the number of YARA alerts, the number of malicious verdicts, and so on. The base score serves as a baseline for prioritizing an IOC.

b) Suspicious Behavior Score: Computed based on the behavioral attributes of a file IOC. Behavioral attributes mainly describe the statistics of suspicious behaviors, such as the number of suspicious commands, the number of IDS alerts, and so on. The suspicious behavior score serves as a secondary score that supports the prioritization of an IOC.

c) Attack Techniques Score: Computed based on the MITRE attack techniques and tactics used by a file IOC. Each attack technique and tactic is assigned a score based on a predefined impact score matrix using the MITRE ATT&CK Navigator. The attack techniques score serves as a secondary score that supports the prioritization of an IOC.

2) IOC Prioritization: In substep (C1), after calculating the three score categories, various prioritization methods are applied to the crawled IOC dataset. A higher score intuitively indicates a higher priority for allocating resources to examine and track down a specific IOC. The use of three different score categories allows us to conduct diverse experiments and produce various prioritization results, which will be presented in Section V.

IV. Methodology and Implementation

A. Data Attributes

In this subsection, we will introduce the extracted VirusTotal IOC data attributes and explain how they are quantified. We will also present the implementation of `FOCUS` that uses the quantified attributes as input parameters.

1) Basic Attributes: The base score of an IOC is calculated using the following fundamental attributes, which are presented in Table I along with their respective data format. The quantification mechanism for each attribute is explained in Table II.

TABLE I: VirusTotal File IOC basic attributes

Attribute Name	Data Format
YARA alerts	[{”rule_name”: ”WannaCry_Ransomware”, ...}, ...]
SIGMA analysis statistics	{”high”: 0, ”medium”: 1, ”critical”: 0, ”low”: 0}
Popular threat classifications	[{“count”: 17, “value”: “trojan”}, ...]
Sandbox verdicts	[{”C2AE”: {”category”: ”malicious”, ”confidence”: 70, ...}, ...]
VirusTotal analysis statistics	{”suspicious”: 0, ”malicious”: 63, ...}
VirusTotal last analysis date	1666399891

TABLE II: Basic attributes quantification mechanism

Attribute Name	Quantification Mechanism	Notation
YARA alerts	Number of YARA alerts	X_1
SIGMA analysis statistics	Number of SIGMA alerts	X_2
Popular threat classifications	Sum of different threat classification counts	X_3
Sandbox verdicts	Average confidence level of sandbox verdicts	X_4
VirusTotal analysis statistics	Number of malicious analysis results	X_5
VirusTotal last analysis date	Number of days since last analysis date	X_6

2) File Behavior Attributes: The suspicious behavior score of an IOC is calculated using the following file behavior attributes, which are presented in Table III along with their respective data format. The quantification mechanism for each attribute is explained in Table IV. The list of suspicious commands and command flags that are used to quantify the command executions attribute are shown in Table V.

TABLE III: VirusTotal File IOC file behavior attributes

Attribute Name	Data Format
IDS alerts	[{"alert_severity": "high", ...}, ...]
Command executions	["su", "ping -c ...", ..]
Registry keys set	[{"key": "HKEY_CURRENT_USER\...", "value": "09 04"}, ...]

TABLE IV: File behavior attributes quantification mechanism

Attribute Name	Quantification Mechanism	Notation
IDS alerts	Number of IDS alerts	Y_1
Command executions	Number of suspicious commands and command flags using Table V	Y_2
Registry keys set	Number of registry keys set	Y_3

TABLE V: Suspicious commands and command flags

cmd	reg	echo	curl	python
bypass	create	su	bash	useradd
net	netsh	wget	sc	findstr
powershell				

3) Attack Techniques: The attack techniques score of an IOC is calculated using the following attack techniques attribute, which is presented in Table VI along with its respective data format. The quantification mechanism for the attribute is explained in Table VII.

TABLE VI: VirusTotal File IOC attack techniques attribute

Attribute Name	Data Format
Attack techniques	[{"technique_id": "T1036", "technique_name": "Masquerading"}, ...]

TABLE VII: Attack techniques attribute quantification mechanism

Attribute Name	Quantification Mechanism	Notation
Attack techniques	Sum of assigned scores of individual attack technique used	Z_n

B. Scoring Mechanism

The following sections introduce three score categories and their scoring equations, which will be used to generate the prioritization experiment results presented in Section V.

1) Base Score:

$$Base\ Score = \sum_{i=1}^{5} X_i \cdot (1 - (\frac{X_6}{ioc\ end\ time})^{\frac{1}{decay\ rate}})$$

where:

X_i : Quantified basic attributes.

$ioc\ end\ time$: The duration in days for an IOC to have a negative decayed base score. In our work, a value of 365 days is used as the IOC end time for all the conducted experiments.

$decay\ rate$: The decaying rate of the base score. In our work, a value of 0.5 is used as the decay rate for all the conducted experiments.

2) Suspicious Behavior Score:

$$Suspicious\ Behavior\ Score = \sum_{i=1}^{3} Y_i$$

where Y_1, Y_2, Y_3 are the quantified file behavior attributes.

3) Attack Techniques Score:

$$Attack\ Techniques\ Score = \sum_{i=1}^{n} Z_i$$

where $Z_1, Z_2, ..., Z_n$ are the quantified attack techniques.

4) Overall Prioritization Score:

$$Overall\ Prioritization\ Score = Rescaled\ Base\ Score + Suspicious\ Behavior\ Score + Attack\ Techniques\ Score$$

where the computed values of base score for each IOC are rescaled to the interval of [0, 100].

V. Performance Evaluation

A. Dataset

The experiments conducted in our work utilize various data sources of IOC, as shown in Table VIII. By utilizing different data sources, we can obtain diverse experiment results and make meaningful comparisons, which will be presented in the subsequent sections.

TABLE VIII: Data source of IOC

Data Source	Amount
Threatfox	1184
ESET Research Group	184
Python DataSynthesizer Synthetic Data	10000

B. Experiments

Three main experiments are conducted in our work to illustrate the pattern and prioritization results of FOCUS. The first experiment investigates the relationship between the various extracted attributes that are used to compute different score categories. The second and third experiments generate prioritization ranking results for the IOC dataset using different score categories.

a) : In all experiments, IOCs with a negative base score are excluded because they are deemed too old to be prioritized in a given IOC dataset. The final amount of data used in the conducted experiments are shown in Table IX

TABLE IX: Amount of IOC with negative base score

Data Source	Original Amount	IOC with Negative Base Score	Final Amount
Threatfox	1184	328	856
ESET Research Group	184	84	100
Python DataSynthesizer Synthetic Data	10000	0	10000

1) Relationship between extracted attributes and score categories: To comprehend how the extracted attributes impact the three FOCUS score categories (described in subsection IV-B), we've generated scatter plots. Each plot illustrates the relationship between an IOC attribute, quantified on the horizontal axis, and a score category on the vertical axis. This experiment employs both the Threatfox IOC dataset and synthetic data, focusing only on the attributes with significant correlations.

a) : As shown in Figure 2, there is an upward trend indicating a positive correlation between the number of malicious analysis results and the base score of an IOC. It is observed that the upper bound of the base score increases with the number of malicious analysis results.

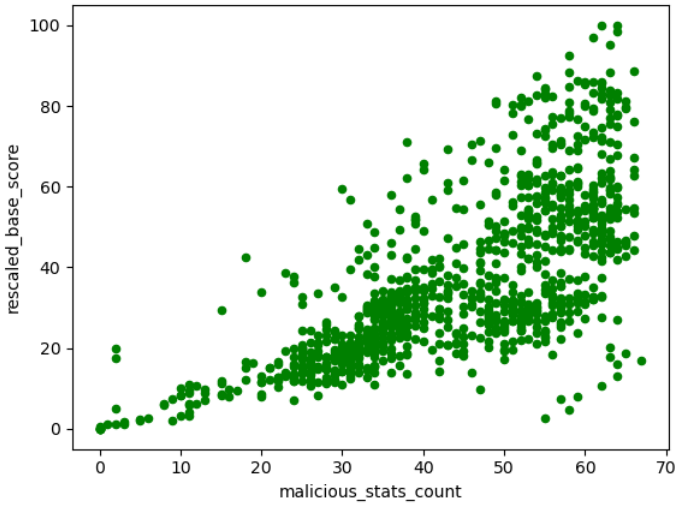
b) : As shown in Figure 3, there is a downward trend indicating a negative correlation between the age of the IOC in days and the base score of an IOC. It is observed that the upper bound of the base score decreases with the increase in the age of the IOC in days.

c) : As shown in Figure 4, there is an upward trend that implies a positive correlation between the number of IDS alerts and the overall prioritization score of an IOC.

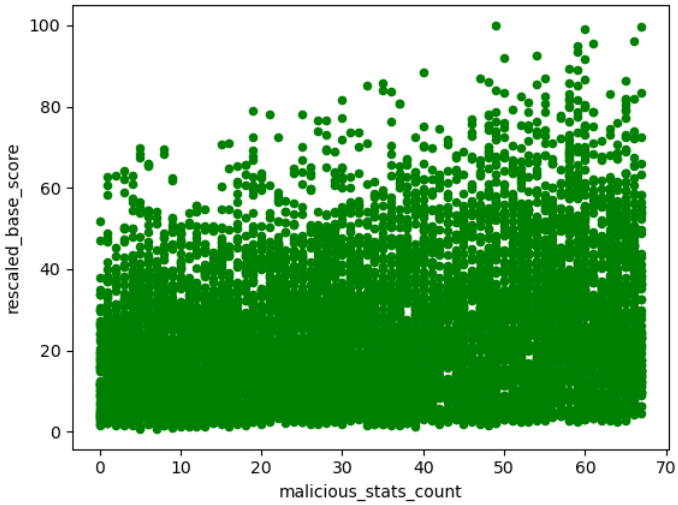
2) One-Phase Prioritization: One-phase prioritization is a ranking method that generates prioritization rankings based on the computed overall prioritization score for each IOC in the scored dataset. In this experiment, only the Threatfox IOC data is used, and several kernel density estimate (KDE) plots are created to visualize the distribution of the quantified IOC attributes for the top 100 prioritized IOCs.

a) : By observing the plotted KDE plots, the top 100 prioritized IOCs have the following characteristics:

1) **Relatively higher** number of YARA alerts.
2) **Relatively higher** number of SIGMA alerts.
3) **Relatively higher** sum of threat classification counts.
4) **Relatively higher** average confidence level of sandbox verdicts.
5) **Relatively higher** number of malicious analysis results.
6) **Relatively lower** age (number of days since last analysis date).



(a) Threatfox IOC



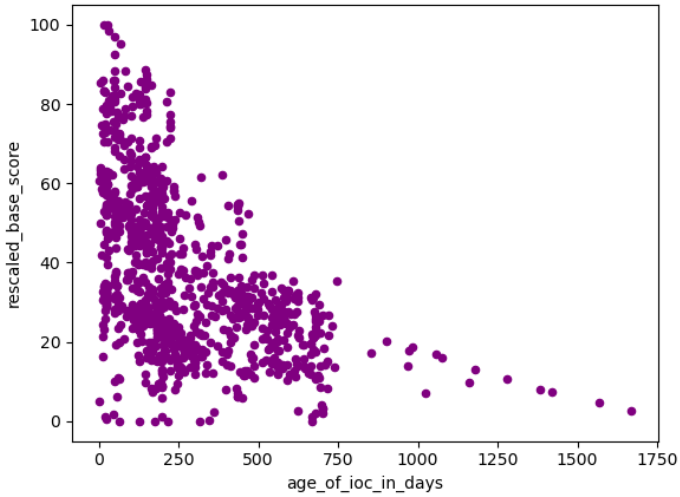
(b) Synthetic Data

Fig. 2: Relationship between malicious analysis results (X_5) and base score

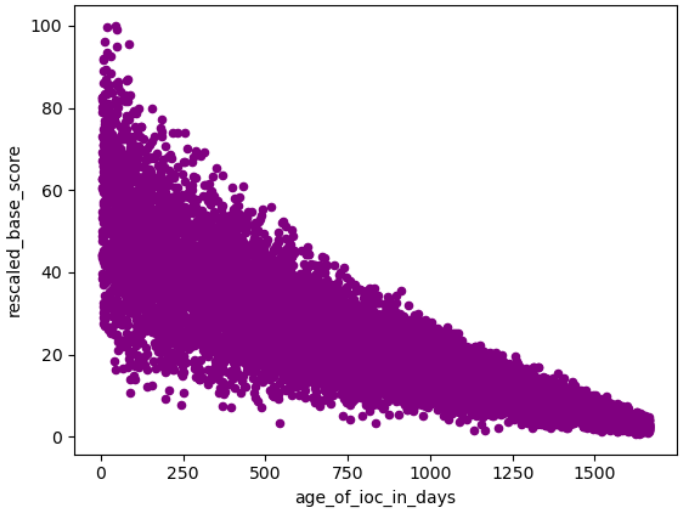
7) **Relatively higher** number of IDS alerts.
8) **Relatively higher** number of suspicious commands.
9) **Relatively higher** number of registry keys set.
10) **Relatively higher** number of sum of assigned scores of attack techniques used.

b) : This observation highlights the prioritization patterns and characteristics of FOCUS when using the one-phase prioritization method.

3) Two-Phase Prioritization: Two-phase prioritization is a method that produces the prioritization rankings of a scored IOC dataset by conducting two prioritization phases. The first phase uses the rescaled base score to establish four severity-

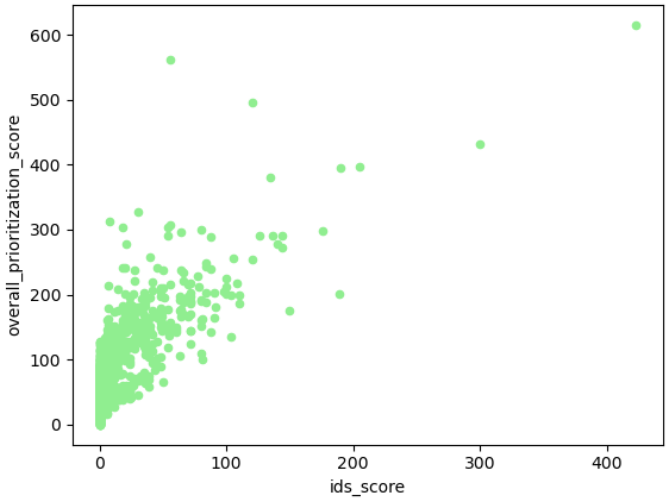


(a) Threatfox IOC

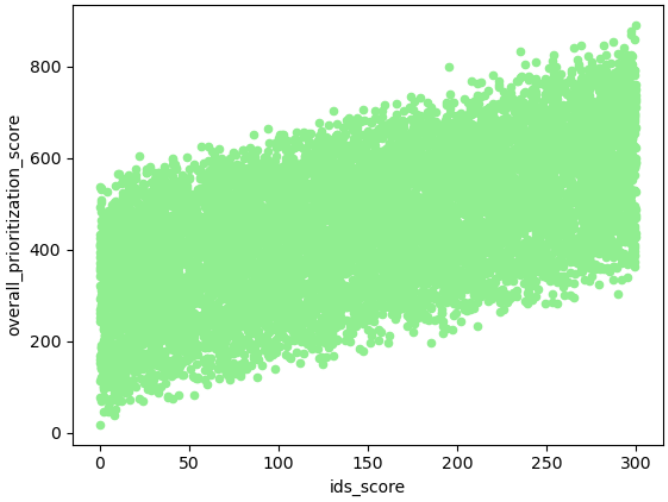


(b) Synthetic Data

Fig. 3: Relationship between age of IOC in days (X_6) and base score



(a) Threatfox IOC



(b) Synthetic Data

Fig. 4: Relationship between number of IDS alerts (Y_1) and overall prioritization score

based prioritization levels for the given IOC dataset. In the second phase, the sum of suspicious behavior score and attack techniques score is used to determine the final prioritization rankings. A detailed discussion of the findings and results of this experiment will be presented in subsection VI-A1.

VI. Results and Discussion

In this section, we present and discuss FOCUS's results and key discussion points.

A. Results

1) One-Phase vs Two-Phase Prioritization: In the first phase of the two-phase prioritization method, IOCs are categorized into four prioritization severity levels: low, medium, high, and critical, based on their rescaled base score. The results of this phase are presented in Table X, where X_i represents the rescaled base score of an IOC. The distribution of IOCs across the severity levels is fairly even, with the critical severity level containing only 64 out of 856 IOCs.

TABLE X: First phase prioritization severity distribution

Prioritization Severity	Rescaled Base Score	Amount
Low	$0 < X_i <= 25$	290
Medium	$25 < X_i <= 50$	302
High	$50 < X_i <= 75$	200
Critical	$75 < X_i$	64
	Total	856

a) : After the second phase of two-phase prioritization, the final rankings for the IOCs are determined based on the sum

of the suspicious behavior score and attack techniques score. Comparing the results of two-phase and one-phase prioritization (Table XI), we observe changes in prioritization rankings. Among the 856 IOCs, 448 show an increase in ranking during the second phase, while 408 experience a decrease. The rest of the IOCs maintain their positions in both one-phase and two-phase prioritization.

TABLE XI: Two-phase vs one-phase prioritization

Difference in rankings	Amount
Increased	448
Decreased	408

To better grasp the impact of two-phase prioritization compared to one-phase prioritization, we've created a stacked bar plot (Figure 5) in the appendix. This plot illustrates how rankings change within different prioritization severity categories, showing the distribution of IOCs with increased or decreased rankings.

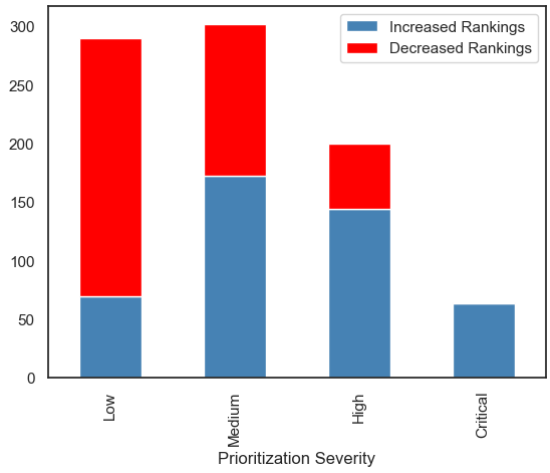


Fig. 5: Two-phase prioritization severity distribution

The stacked bar plot results show that two-phase prioritization lowered rankings for numerous low-priority IOCs but enhanced rankings for many high and critical ones. This trend primarily affected the higher-priority categories, with only a minor impact on low and medium-priority IOCs.

B. Discussion

Q1 **What is the significance of addressing IOC data fatigue, and how does FOCUS offer a solution to this problem?** - Managing IOC data fatigue is crucial to prevent analysts from being overwhelmed and missing critical alerts, ultimately improving incident response efficiency. FOCUS enables customized IOC prioritization, allowing users to focus on the base score while excluding attack technique scores.

Q2 **What is the purpose of using a sythentic data in the conducted experiments?** - Our work utilizes the Python DataSynthesizer data generator to simulate and supplement the provided dataset, mitigating the scarcity of Threatfox IOC data. This approach enhances the analysis outcomes by incorporating synthetic data.

Q3 **To what extent does the effectiveness of FOCUS depend on the accuracy of the analysis performed by VirusTotal's scanning engine?** - FOCUS assesses IOC priority based on VirusTotal IOC attributes. Its effectiveness is correlated with VirusTotal's scanning engine accuracy, yet it provides adaptability and alternative attributes to mitigate inaccuracies.

VII. Conclusion

In this study, our goal was to create a scoring mechanism for efficient IOC data prioritization and resource allocation. We utilized VirusTotal file IOC attributes to quantify parameters and assess their correlation with score categories. By employing two prioritization methods and comparing their outcomes, we successfully obtained prioritization scores and rankings for the IOC data. Future research could involve optimizing hyperparameters for enhanced effectiveness. Additionally, we can explore prior work on IoC clustering to refine IoC prioritization and advance Threat Intelligence generation.

Acknowledgement

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